

Random-Key Metaheuristic and Linearization for the Quadratic Multiple Constraints Variable-Sized Bin Packing Problem

1. Gurobi results from original and linearized models

Table 1: Gurobi results

Instance		Original model			Linearized model		
ID	n	LB	Best	Gap(%)	LB	Best	Gap(%)
1	25	6519	7253	10.1	7014	7253	3.3
2	25	6557	6968	5.9	6968	6968	0.0
3	25	6438	7022	8.3	6758	6935	2.6
4	25	9437	11371	17.0	9979	11415	12.6
5	25	9023	11082	18.6	9199	11304	18.6
6	25	8919	11123	19.8	9087	11211	19.0
7	25	4976	5125	2.9	5125	5125	0.0
8	25	4484	4967	9.7	4967	4967	0.0
9	25	4385	4965	11.7	4965	4965	0.0
10	25	6322	7713	18.0	6685	7713	13.3
11	25	5689	7603	25.2	6262	7679	18.4
12	25	5960	7176	17.0	6134	7441	17.6
13	25	8868	10787	17.8	9353	10787	13.3
14	25	8390	10443	19.7	8881	10695	17.0
15	25	8597	10185	15.6	9101	10463	13.0
16	25	15276	17990	15.1	16057	18021	10.9
17	25	14364	17756	19.1	14898	17897	16.8
18	25	13268	17308	23.3	13608	17093	20.4
19	25	6674	8171	18.3	7258	7949	8.7
20	25	7041	7748	9.1	7376	7689	4.1
21	25	6608	7803	15.3	7370	7615	3.2
22	25	9123	11361	19.7	9678	11521	16.0
23	25	8896	11274	21.1	9139	11130	17.9
24	25	8764	11003	20.4	8999	11330	20.6
25	50	10359	25237	59.0	9595	25616	62.5
26	50	9406	24922	62.3	9406	24455	61.5

Instance		Original model			Linearized model		
ID	n	LB	Best	Gap(%)	LB	Best	Gap(%)
27	50	9950	24259	59.0	9434	23951	60.6
28	50	14451	33174	56.4	14607	32491	55.0
29	50	14661	32915	55.5	14476	33000	56.1
30	50	14568	32203	54.8	14791	32863	55.0
31	50	6344	20988	69.8	6488	20966	69.0
32	50	5566	18854	70.5	5821	19493	70.1
33	50	5608	18876	70.3	6423	19585	67.2
34	50	9440	25377	62.8	10192	25535	60.1
35	50	8614	25977	66.8	9250	25984	64.4
36	50	8218	24771	66.8	8858	25571	65.4
37	50	15323	31790	51.8	14942	31508	52.6
38	50	15323	31505	51.4	14853	31603	53.0
39	50	14984	31299	52.1	14902	31355	52.5
40	50	24973	45695	45.4	25085	45507	44.9
41	50	24434	45819	46.7	23923	45670	47.6
42	50	22912	45103	49.2	22617	43860	48.4
43	50	10814	26210	58.7	10719	26703	59.9
44	50	10426	25813	59.6	10919	26148	58.2
45	50	11238	25647	56.2	10375	26294	60.5
46	50	14475	32768	55.8	15175	34930	56.6
47	50	14973	32833	54.4	14161	33205	57.4
48	50	14630	32316	54.7	14267	33651	57.6
49	100	17153	100850	83.0	17217	100173	82.8
50	100	17120	105832	83.8	17115	104678	83.6
51	100	17111	102547	83.3	17110	104330	83.6
52	100	27223	124056	78.1	27533	125941	78.1
53	100	27237	126271	78.4	27190	129477	79.0
54	100	27189	136308	80.0	27185	131648	79.4
55	100	10218	95354	89.3	10154	95298	89.4
56	100	9185	89120	89.7	9571	88579	89.2
57	100	9271	89392	89.6	9296	87901	89.4
58	100	16074	111149	85.5	15755	112180	86.0
59	100	15577	108556	85.6	15631	108368	85.6
60	100	14859	108521	86.3	15104	108445	86.1
61	100	26205	111705	76.5	26169	111116	76.4
62	100	25400	111573	77.2	26135	111494	76.6
63	100	24847	110730	77.6	25066	112157	77.6
64	100	46806	152214	69.2	44932	148884	69.8
65	100	45547	146962	69.0	45595	146146	68.8
66	100	42899	142256	69.8	43378	144898	70.1
67	100	17880	101135	82.3	18905	100798	81.2
68	100	17860	100817	82.3	18121	103431	82.5
69	100	17707	102307	82.7	17810	102549	82.6

Instance		Original model			Linearized model		
ID	n	LB	Best	Gap(%)	LB	Best	Gap(%)
70	100	26710	120139	77.8	26541	118839	77.7
71	100	26461	117592	77.5	26570	116803	77.2
72	100	26268	115080	77.2	26330	116717	77.4
73	200	30470	375104	91.9	30470	393151	92.2
74	200	30470	374153	91.9	30470	374507	91.9
75	200	30470	375918	91.9	30470	376089	91.9
76	200	50565	409442	87.6	50565	443658	88.6
77	200	50565	415919	87.8	50565	457660	89.0
78	200	50565	429660	88.2	50565	456602	88.9
79	200	16107	360694	95.5	16107	375502	95.7
80	200	14459	350099	95.9	14460	376069	96.2
81	200	14442	344132	95.8	14442	377639	96.2
82	200	27553	382758	92.8	27553	405994	93.2
83	200	26310	387238	93.2	26310	414551	93.6
84	200	25987	384555	93.2	25987	435637	94.0
85	200	45272	393751	88.5	45272	424766	89.3
86	200	45229	394673	88.5	45229	428353	89.4
87	200	44800	394898	88.7	44800	429283	89.6
88	200	85027	457451	81.4	85028	495363	82.8
89	200	85027	459525	81.5	85028	539772	84.2
90	200	79786	464271	82.8	79786	581736	86.3
91	200	31556	373626	91.6	31557	395750	92.0
92	200	31556	374781	91.6	31557	391674	91.9
93	200	31490	371865	91.5	31491	379347	91.7
94	200	49357	408022	87.9	49358	438879	88.8
95	200	49196	406270	87.9	49196	444231	88.9
96	200	48657	412571	88.2	48658	484192	90.0

2. RKO results

Table 2: RKO results

ID	n	c	d	m	LB	BKS	RKO	Time(s)	Gap	BRPD	ARPD
1	25	B1	3	10	7014	7253	7253	0.0	3.3	0.0	0.0
2	25	B1	3	20	6968	6968	6968	0.0	0.0	0.0	0.0
3	25	B1	3	50	6758	6935	6935	0.5	2.6	0.0	0.0
4	25	B1	5	10	9979	11371	11371	15.1	12.2	0.0	0.0
5	25	B1	5	20	9199	11082	10978	26.7	16.2	-0.9	-0.7
6	25	B1	5	50	9087	11123	10855	23.0	16.3	-2.5	-1.9
7	25	B2	3	10	5125	5125	5125	0.0	0.0	0.0	0.0
8	25	B2	3	20	4967	4967	4967	0.0	0.0	0.0	0.0
9	25	B2	3	50	4965	4965	4965	0.0	0.0	0.0	0.0

ID	n	c	d	m	LB	BKS	RKO	Time(s)	Gap	BRPD	ARPD
10	25	B2	5	10	6685	7713	7713	0.2	13.3	0.0	0.0
11	25	B2	5	20	6262	7603	7299	3.3	14.2	-4.2	-4.0
12	25	B2	5	50	6134	7176	7115	0.1	13.8	-0.9	-0.5
13	25	B3	3	10	9353	10787	10787	0.7	13.3	0.0	0.0
14	25	B3	3	20	8881	10443	10276	0.6	13.6	-1.6	-1.6
15	25	B3	3	50	9101	10185	10185	0.8	10.6	0.0	0.6
16	25	B3	5	10	16057	17990	17630	4.2	8.9	-2.0	-1.0
17	25	B3	5	20	14898	17756	17401	0.4	14.4	-2.0	-0.8
18	25	B3	5	50	13608	17093	16489	2.6	17.5	-3.7	-1.6
19	25	B4	3	10	7258	7949	7949	1.3	8.7	0.0	0.2
20	25	B4	3	20	7376	7689	7689	0.0	4.1	0.0	0.0
21	25	B4	3	50	7370	7615	7615	0.6	3.2	0.0	0.1
22	25	B4	5	10	9678	11361	11125	0.8	13.0	-2.1	-1.9
23	25	B4	5	20	9139	11130	10779	0.6	15.2	-3.3	-3.0
24	25	B4	5	50	8999	11003	10608	7.8	15.2	-3.7	-3.2
25	50	B1	3	10	10359	25237	24117	6.1	57.0	-4.6	-4.1
26	50	B1	3	20	9406	24455	22783	6.3	58.7	-7.3	-6.8
27	50	B1	3	50	9950	23951	22627	16.2	56.0	-5.9	-4.7
28	50	B1	5	10	14607	32491	31351	14.0	53.4	-3.6	-0.7
29	50	B1	5	20	14661	32915	31908	6.7	54.1	-3.2	-2.3
30	50	B1	5	50	14791	32203	31127	47.6	52.5	-3.5	-2.0
31	50	B2	3	10	6488	20966	20244	22.2	68.0	-3.6	-2.9
32	50	B2	3	20	5821	18854	18449	3.7	68.4	-2.2	-1.6
33	50	B2	3	50	6423	18876	18449	3.0	65.2	-2.3	-1.8
34	50	B2	5	10	10192	25377	25067	12.3	59.3	-1.2	-0.4
35	50	B2	5	20	9250	25977	24777	19.2	62.7	-4.8	-3.5
36	50	B2	5	50	8858	24771	24117	25.0	63.3	-2.7	-1.7
37	50	B3	3	10	15323	31508	30858	35.2	50.3	-2.1	-0.8
38	50	B3	3	20	15323	31505	30932	7.8	50.5	-1.9	-0.5
39	50	B3	3	50	14984	31299	30672	25.4	51.1	-2.0	-0.7
40	50	B3	5	10	25085	45507	44289	35.8	43.4	-2.8	-1.3
41	50	B3	5	20	24434	45670	44806	38.9	45.5	-1.9	-0.4
42	50	B3	5	50	22912	43860	42848	44.6	46.5	-2.4	0.6
43	50	B4	3	10	10814	26210	25393	7.1	57.4	-3.2	-2.3
44	50	B4	3	20	10919	25813	24574	53.5	55.6	-5.0	-4.2
45	50	B4	3	50	11238	25647	24460	36.9	54.1	-4.9	-4.1
46	50	B4	5	10	15175	32768	31372	40.1	51.6	-4.4	-2.9
47	50	B4	5	20	14973	32833	31658	14.2	52.7	-3.7	-2.2
48	50	B4	5	50	14630	32316	31366	47.3	53.4	-3.0	-1.5
49	100	B1	3	10	17217	100173	93843	148.5	81.7	-6.7	-5.9
50	100	B1	3	20	17120	97933	90707	124.6	81.1	-8.0	-6.9
51	100	B1	3	50	17111	97922	90642	117.3	81.1	-8.0	-7.3
52	100	B1	5	10	27533	116348	110271	135.5	75.0	-5.5	-4.0
53	100	B1	5	20	27237	116494	109971	77.4	75.2	-5.9	-4.6

ID	n	c	d	m	LB	BKS	RKO	Time(s)	Gap	BRPD	ARPD
54	100	B1	5	50	27189	115508	108961	137.4	75.0	-6.0	-5.3
55	100	B2	3	10	10218	92689	85734	120.9	88.1	-8.1	-7.4
56	100	B2	3	20	9571	88579	81464	115.6	88.3	-8.7	-7.7
57	100	B2	3	50	9296	87901	81492	159.7	88.6	-7.9	-7.0
58	100	B2	5	10	16074	102688	95578	77.2	83.2	-7.4	-6.3
59	100	B2	5	20	15631	101898	96129	91.8	83.7	-6.0	-5.0
60	100	B2	5	50	15104	99941	93800	76.6	83.9	-6.5	-5.5
61	100	B3	3	10	26205	111116	108917	68.4	75.9	-2.0	-1.1
62	100	B3	3	20	26135	111494	109436	131.7	76.1	-1.9	-0.9
63	100	B3	3	50	25066	110730	108655	65.5	76.9	-1.9	-0.7
64	100	B3	5	10	46806	141233	136503	82.6	65.7	-3.5	-2.5
65	100	B3	5	20	45595	141348	137609	144.0	66.9	-2.7	-1.7
66	100	B3	5	50	43378	140308	135262	151.1	67.9	-3.7	-2.6
67	100	B4	3	10	18905	100798	96329	81.8	80.4	-4.6	-3.6
68	100	B4	3	20	18121	100817	94064	66.6	80.7	-7.2	-6.1
69	100	B4	3	50	17810	101713	93914	155.7	81.0	-8.3	-7.5
70	100	B4	5	10	26710	115442	108894	48.0	75.5	-6.0	-4.3
71	100	B4	5	20	26570	115037	109064	174.9	75.6	-5.5	-4.3
72	100	B4	5	50	26330	114366	108810	101.6	75.8	-5.1	-4.1
73	200	B1	3	10	30470	375104	363104	218.5	91.6	-3.3	-3.0
74	200	B1	3	20	30470	374153	356945	237.8	91.5	-4.8	-4.4
75	200	B1	3	50	30470	375918	355549	261.0	91.4	-5.7	-5.1
76	200	B1	5	10	50565	409442	397386	239.3	87.3	-3.0	-2.5
77	200	B1	5	20	50565	415919	394957	113.4	87.2	-5.3	-4.3
78	200	B1	5	50	50565	421503	392877	179.9	87.1	-7.3	-6.8
79	200	B2	3	10	16107	360694	346867	325.6	95.4	-4.0	-3.7
80	200	B2	3	20	14460	350099	338312	250.0	95.7	-3.5	-3.1
81	200	B2	3	50	14442	344132	338478	229.3	95.7	-1.7	-1.3
82	200	B2	5	10	27553	382758	368084	163.5	92.5	-4.0	-3.5
83	200	B2	5	20	26310	387238	365656	257.3	92.8	-5.9	-5.4
84	200	B2	5	50	25987	384555	363227	220.2	92.8	-5.9	-5.5
85	200	B3	3	10	45272	393751	393825	249.9	88.5	0.0	0.5
86	200	B3	3	20	45229	394673	394159	196.2	88.5	-0.1	0.6
87	200	B3	3	50	44800	394898	393288	218.3	88.6	-0.4	0.4
88	200	B3	5	10	85028	457451	449657	283.8	81.1	-1.7	-1.3
89	200	B3	5	20	85028	459525	451109	231.0	81.2	-1.9	-1.0
90	200	B3	5	50	79786	464271	446762	167.3	82.1	-3.9	-3.0
91	200	B4	3	10	31557	373626	368836	287.4	91.4	-1.3	-1.1
92	200	B4	3	20	31557	374781	364081	249.2	91.3	-2.9	-2.2
93	200	B4	3	50	31491	371865	364175	345.9	91.4	-2.1	-1.8
94	200	B4	5	10	49358	408022	394708	120.9	87.5	-3.4	-2.7
95	200	B4	5	20	49196	406270	394761	244.5	87.5	-2.9	-2.4
96	200	B4	5	50	48658	412571	394668	244.4	87.7	-4.5	-4.3